

7. ELECTRICALS

1.	AC Motor 1120 KW for 550mm Roughing Mill	1 No.
2.	300 KW D.C. Motor for 400mm Continuous Mill	2 Nos.
3.	300 KW D.C. Motor for 360mm Continuous Mill	4 Nos.
4.	300 KW D.C. Motor for 320mm Continuous Mill	4 Nos.
5.	300 KW D.C. Motor for 280mm Continuous Mill	4 Nos.
6.	Small AC Motors for Y Table, Turning Wall Roller table, Approach Roller table, etc.	1 Set
7.	Main LT A.C. Incomer Panel for all AC / DC Motors	1 Set
8.	D.C Drive Control Panel with Control Desk for Shears & Pinch Rolls	7 Nos.
9.	Scada base DC Drives Control Panel with Control Desk for 400mm, 360mm, 320mm & 280mm Continuous Mill	14 Nos.
10.	PLC, PCs, Remote I/Os etc System for Rolling Mill	1 Set
11.	Motorized Liquid Resistance starter for 1120 KW Motor	1 No.
12.	DC Motors for Shears & Pinch Roll	7 Nos.
13.	ACB for DC Incomer	1 No.
14.	APFC Panel with Capacitors	1 No.

Note : MV Side, Transformer, all Cables & Earthing's are not included



PRELIMINARY AC & DC MOTOR LIST										
MACHINE										
Item No.	Designation	Qty	Rated Each (KW)	Total Connected (KW)	Rated Speed (RPM)	Simultaneous working (KW)	Type of Motor	Voltage (V)	Type of Driven/ Duty	LOCATION
1.00	ROUGHING MILL									
1.01	550 MM 3-Hi Mill Stand	1	1120	1120	750	1120	DC	400	THY-2Q	DRIVE PANEL
1.02	Crop-Cum Cobble Shear -1	1	110	110	0/600	110	DC	400	THY-4Q	DRIVE PANEL
1.03	Lubrication Unit	1	7.5	15	1440	30	IM	415	S / D	MCC#1
2.00	INTERMEDIATE MILL 400mm									
2.01	400 MM 2-Hi Mill Stands	2	300	600	0/750/1250	2100	DC	400	THY-2Q	DRIVE PANEL
2.02	Lubrication Unit	1+1	7.5	15	1440	15	IM	415	S / D	MCC#1
2.03	Crop-Cum Cobble Shear	1	130	130	0/600	130	DC	400	THY-4Q	DRIVE PANEL
3.00	INTERMEDIATE MILL 360mm									
3.01	360 MM 2-Hi Mill Stands	4	300	1200	0/750/1250	2100	DC	400	THY-2Q	DRIVE PANEL
3.02	Lubrication Unit	1+1	7.5	15	1440	15	IM	415	S / D	MCC#1
3.03	Crop-Cum Cobble Shear -2	1	130	130	0/600	130	DC	400	THY-4Q	DRIVE PANEL
4.00	INTERMEDIATE MILL 320mm									
4.01	320 MM 2-Hi Mill Stand	4	300	1200	0/1000/1800	2100	DC	400	THY-2Q	DRIVE PANEL
4.02	Lubrication Unit	1+1	7.5	15	1440	15	IM	415	S / D	MCC#1
5.00	FINISHING MILL 280mm									
5.01	320 MM 2-Hi Mill Stand	4	300	1200	0/1000/1800	2700	DC	400	THY-2Q	DRIVE PANEL
5.02	Lubrication Unit	1+1	7.5	15	1440	15	IM	415	S / D	MCC#1



PRELIMINARY AC & DC MOTOR LIST										
MACHINE										
Item No.	Designation	Qty	Rated Each (KW)	Total Connected (KW)	Rated Speed (RPM)	Simultaneous working (KW)	Type of Motor	Voltage (V)	Type of Driven/Duty	LOCATION
6	STRAIGHT TMT LINE									
6.01	Pre & Post Pinch Roll	2	45	90	0/1800	135	DC	400	THY-2Q	DRIVE PANEL
6.02	Flying Shear Crank Type	1	270	270	0/1000	270	DC	400	THY-2Q	DRIVE PANEL
6.03	Lub Unit For Shears & Pinch Roll	1+1	5.5	5.5	1440	5.5	IM	415	DOL	DRIVE PANEL
6.04	Tail Brake Pinch Roll	2	60	120	0/1800	240	DC	400	THY-2Q	DRIVE PANEL
6.05	Twin Channel	1	15	15	1440	36	SC	415	DOL	DRIVE PANEL
7	COOLING BED (66 M)									
7.01	Aligning Rollers	28	1.1	30.8	110	30.8	GM	415	VVFD	DRIVE PANEL
7.02	Cooling Bed Rake Drive	1	73	73	0/1000	80	DC	400	THY-4Q	DRIVE PANEL
7.03	Layer Former (Fixed Type)	2	7.5	15	21	15	GM	415	VVFD	DRIVE PANEL
7.04	Bar transfer Arrangement in Layers	2	22.5	45	47	45	GM	415	VVFD	DRIVE PANEL
7.05	Run Out Roller Table	58	2.2	127.6	150	127.6	GM	415	VVFD	DRIVE PANEL
8	COLD SHEAR AREA									
8.01	Cold Shear	1	30	30	1440	60	IM	415	S / D	MCC#2
8.02	Lubrication Unit for Cold Shear	1	1.5	1.5	1440	1.5	IM	415	DOL	MCC#2
8.03	Cold Shear Run Out Roller Table (Depressing Table) – 6 Mtr	5	2.2	11	150	11	GM	415	VVFD	DRIVE PANEL
9	MATERIAL HANDLING AREA									
9.01	Chain Transfer Run In Roller Table	21	2.2	46.2	150	46.2	GM	415	VVFD	DRIVE PANEL
9.02	Chain Transfer No.-1	2	11	22	75	22	GM	415	VVFD	DRIVE PANEL
9.03	Chain Transfer No.-2	2	7.5	15	75	15	GM	415	VVFD	DRIVE PANEL
9.04	Bundle Forming Devices	1	7.5	7.5	75	7.5	SC	380	VVFD	DRIVE PANEL
9.05	Roller Table below Bundle Former	11	2.2	24.2	150	24.2	GM	415	VVFD	DRIVE PANEL
9.06	Hyd. Power Pack for Bar Handling Area	1+1	37	74	1440	74	IM	415	S/D	MCC#2
9.07	Tying M/c	1	13KVA			0			FEEDER	DRIVE PANEL
9.08	Tying Machine Run In Roller Table	11	2.2	24.2	150	24.2	GM	415	VVFD	DRIVE PANEL
9.09	Tied bundle Collection Roller Table	11	2.2	24.2	150	24.2	GM	415	VVFD	DRIVE PANEL
9.10	Bundle Storage Device	2	15	30	960	30	IM	415	VVFD	DRIVE PANEL



AC MOTOR TECHNICAL DETAIL 1120 KW		
SL.	PARTICULARS	
1	Type	WOUND
2	Mounting (IM)	1001
3	Index of Protection (IP)	23
4	Index of Cooling (IC)	01
5	Rating (kW)	1125
6	Polarity	8
7	Frame	KCW560C3
8	Rated Voltage(V)	415 +/- 10%
9	Frequency(Hz)	50 +/- 5%
10	Rated current (A)	1992
11	Rated Speed (RPM)	741
12	Rotor Voltage (V)	1120
13	Rotor Current (A)	620
14	Pull Out Torque (P.U)	2.20
15	Percentage Efficiency	
	Full load	95.8
	$\frac{3}{4}$ FL	95.5
	$\frac{1}{2}$ FL	94.0
16	Power Factor	
	Full Load	0.82
	$\frac{3}{4}$ FL	0.78
	$\frac{1}{2}$ FL	0.70
17	Insulation Class Stator/ Rotor	F/F
18	Temperature rise Rotor/ Stator(°C)	95/95
19	Method of Starting	LRS
20	Duty	S1
21	Amb Temperature (°C)	50
22	Noise & Vibration Limit	AS PER IS 325
23	Drive transmission	DIRECT FLEXIBLE



Sl No	Description	Qty
1	<u>DC Variable speed drive for 350 Kw , 450 VDC, 800 Amp DC motor for Rolling Stand</u> All the drives and switchgears are suitably rated to take care of Inrush current of 200 % for period of 10 Sec . Field Supply is integrated in the Drive module. Drives are of CLASS IV overload capacity.	14 Set
A	- Parker DC Variable Speed , 599 series- Digital Microprocessor based Thyristor DC Drive of 1050 Amp with Class 4 Overload , suitable for 350 KW, 2Q DC Motor for stands. The Drive operates in closed loop, Speed control mode, to control Speed , Motor Torque, Armature Voltage and Field Current. - 630 Amp MCCB for control and Isolation of each DC Drive. - Input Ac side Semi-Conductor Fuses and line Reactor for Protection. - Encoder interface card for pulse encoder - Profibus interface card mounted on the drive module for High Speed communication with drive/ PLC. - Block Programming with Readymade function Block to do Co-Ordinated Drive Control. - 16 Character, 2 Line MMI Display – Keypad unit per drive , mounted on the front door of the drive panel for for drive programming and diagnostics. - DOL Starter through MPCB for the Blower fan of Motor. - All the above enclosed in 2 mm CRCA steel Sheet enclosure having IP 31 Protection , with Aluminum Busbars. - Switchgear of Siemens/Schneider/L & T Make	14 Sets

Sl No	Description	Qty
2	<u>DC Variable speed drive for 80 Kw , 440 VDC, 200 Amp DC motor for CC Shear-1</u> All the drives and switchgears are suitably rated to take care of Inrush current of 200 % for period of 10 Sec . Field Supply is integrated in the Drive module. Drives are of CLASS IV overload capacity.	1 Set
a	- Parker DC Variable Speed , 598 series- Digital Microprocessor based Thyristor DC Drive of 400 Amp with Class 4 Overload , suitable for 80 KW, 2Q DC Motor for CC Shear. The Drive operates in closed loop, Speed control mode, to control Speed , Motor Torque, Armature Voltage and Field Current. - 1250 Amp MCCB for control and Isolation of each DC Drive. - Input Ac side Semi-Conductor Fuses and line Reactor for Protection.- - Encoder interface card for pulse encoder - Profibus interface card mounted on the drive module for High Speed communication with drive/ PLC. - Block Programming with Readymade function Block to do Co-Ordinated Drive Control. - 16 Character, 2 Line MMI Display – Keypad unit per drive , mounted on the front door of the drive panel for for drive programming and diagnostics. - DOL Starter through MPCB for the Blower fan of Motor.- - All the above enclosed in 2 mm CRCA steel Sheet enclosure having IP 31 Protection , with Aluminum Busbars. - Switchgear of Siemens/Schneider/L & T Make	1 Set

Sl No	Description	Qty
3	<u>DC Variable speed drive for 130 Kw , 440 VDC, 200 Amp DC motor for CC Shear-2</u> All the drives and switchgears are suitably rated to take care of Inrush current of 200 % for period of 10 Sec . Field Supply is integrated in the Drive module. Drives are of CLASS IV overload capacity.	1 Set
a	- Parker DC Variable Speed , 598 series- Digital Microprocessor based Thyristor DC Drive of 400 Amp with Class 4 Overload , suitable for 80 KW, 2Q DC Motor for CC Shear. The Drive operates in closed loop, Speed control mode, to control Speed , Motor Torque, Armature Voltage and Field Current. - 1250 Amp MCCB for control and Isolation of each DC Drive. - Input Ac side Semi-Conductor Fuses and line Reactor for Protection.- - Encoder interface card for pulse encoder - Profibus interface card mounted on the drive module for High Speed communication with drive/ PLC. - Block Programming with Readymade function Block to do Co-Ordinated Drive Control. - 16 Character, 2 Line MMI Display – Keypad unit per drive , mounted on the front door of the drive panel for for drive programming and diagnostics. - DOL Starter through MPCB for the Blower fan of Motor.- - All the above enclosed in 2 mm CRCA steel Sheet enclosure having IP 31 Protection , with Aluminum Busbars. - Switchgear of Siemens/Schneider/L & T Make	1 Set

4. DC Variable speed drive for 89 Kw , 440 VDC, 190 Amp DC motor for Pinch roll

All the drives and switchgears are suitably rated to take care of Inrush current of 200 % for period of 10 Sec . Field Supply is integrated in the Drive module
Drives are of CLASS IV overload capacity.

A	- Parker DC Variable Speed , 591 series- Digital Microprocessor based Thyristor DC Drive of 270 Amp Continuous Current, suitable for 89 KW, 2QDC Motor for Pinch roll . The Drive operates in closed loop, Speed control mode, to control Speed , Motor Torque, Armature Voltage and Field Current. - 160 Amp MCCB for control and Isolation of each DC Drive. - Input Ac side Semi-Conductor Fuses and line Reactor for Protection. - Encoder interface card for pulse encoder - Profibus interface card mounted on the drive module for High Speed communication with drive/ PLC. - Block Programming with Readymade function Block to do Co-Ordinated Drive Control. - 16 Character, 2 Line MMI Display – Keypad unit per drive , mounted on the front door of the drive panel for for drive programming and diagnostics. - DOL Starter through MPCB for the Blower fan of Motor.- - All the above enclosed in 2 mm CRCA steel Sheet enclosure having IP 31 Protection , with Aluminum Busbars. - Switchgear of Siemens/Schneider/L & T Make	2 Sets
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4. DC Variable speed drive for 89 Kw , 440 VDC, 190 Amp DC motor for TB**- 2 Set**

All the drives and switchgears are suitably rated to take care of Inrush current of 200 % for period of 10 Sec . Field Supply is integrated in the Drive module.

Drives are of CLASS IV overload capacity

a	- Parker DC Variable Speed , 590 series- Digital Microprocessor based Thyristor DC Drive of 270 Amp Continuous Current, suitable for 89 KW, 4QDC Motor for TB . The Drive operates in closed loop, Speed control mode, to control Speed , Motor Torque, Armature Voltage and Field Current. - 200 Amp MCCB for control and Isolation of each DC Drive. - Input Ac side Semi-Conductor Fuses and line Reactor for Protection. - Encoder interface card for pulse encoder - Profibus interface card mounted on the drive module for High Speed communication with drive/ PLC. - Block Programming with Readymade function Block to do Co-Ordinated Drive Control. - 16 Character, 2 Line MMI Display – Keypad unit per drive , mounted on the front door of the drive panel for for drive programming and diagnostics. - DOL Starter through MPCB for the Blower fan of Motor. - All the above enclosed in 2 mm CRCA steel Sheet enclosure having IP 31 Protection , with Aluminum Busbars. - Switchgear of Siemens/Schneider/L & T Make	2 Sets
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2) DC Variable speed drive for 225 Kw , 440 VDC, 600 Amp DC motor for TMT Shear 1 Set

All the rives and switchgears are suitably rated to take care of Inrush current of 200 % for period of 10 Sec. Field Supply is integrated in the Drive module. Drives are of CLASS IV overload capacity

A	- Parker DC Variable Speed , 598 series- Digital Microprocessor based Thyristor DC Drive of 1250 Amp With Class 4 Overload , suitable for 225 KW, 2QDC Motor for TB . The Drive operates in closed loop, Speed control mode, to control Speed , Motor Torque, Armature Voltage and Field Current. - 800 Amp MCCB for control and Isolation of each DC Drive. - Input Ac side Semi-Conductor Fuses and line Reactor for Protection. - Encoder interface card for pulse encoder - Profibus interface card mounted on the drive module for High Speed communication with drive/ PLC. - Block Programming with Readymade function Block to do Co-Ordinated Drive Control. - 16 Character, 2 Line MMI Display – Keypad unit per drive , mounted on the front door of the drive panel for for drive programming and diagnostics. - DOL Starter through MPCB for the Blower fan of Motor. - All the above enclosed in 2 mm CRCA steel Sheet enclosure having IP 31 Protection , with Aluminum Busbars. - Switchgear of Siemens/Schneider/L & T Make	1 Sets
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6 DC Variable speed drive for 55 Kw , 440 VDC, 130 Amp DC motor**- 1 Set**

For Cont Shear All the drives and switchgears are suitably rated to take care of Inrush current of 200 % for period of 10 Sec . Field Supply is integrated in the Drive module. Drives are of CLASS IV overload capacity.

A	- Parker DC Variable Speed , 590 series- Digital Microprocessor based Thyristor DC Drive of 270 Amp Continuous Current, suitable for 55 KW, 4QDC Motor for Continuous shear. The Drive operates in closed loop, Speed control mode, to control Speed , Motor Torque, Armature Voltage and Field Current. - 200 Amp MCCB for control and Isolation of each DC Drive - Input Ac side Semi-Conductor Fuses and line Reactor for Protection. - Encoder interface card for pulse encoder - Profibus interface card mounted on the drive module for High Speed communication with drive/ PLC. - Block Programming with Readymade function Block to do Co-Ordinated Drive Control. - 16 Character, 2 Line MMI Display – Keypad unit per drive , mounted on the front door of the drive panel for for drive programming and diagnostics. - DOL Starter through MPCB for the Blower fan of Motor. - All the above enclosed in 2 mm CRCA steel Sheet enclosure having IP 31 Protection , with Aluminum Busbars.	1 Sets
B	- Switchgear of Siemens/Schneider/L & T Make - Mitsubishi servo drive for shifter - Mitsubishi , Servo Motor - Gearhead - With Motor power and resolver feedback cable	

7. DC Variable speed drive for 15 Kw , 440 VDC, 40 Amp DC motor for Twin Channel**1-Set**

All the drives and switchgears are suitably rated to take care of Inrush current of 200 % for period of 10 Sec. Field Supply is integrated in the Drive module Drives are of CLASS IV overload capacity.

A	- Parker DC Variable Speed , 590 series- Digital Microprocessor based Thyristor DC Drive of 110 Amp Continuous Current, suitable for 15 KW, 4QDC Motor for Twin Channel . The Drive operates in closed loop, Speed control mode, to control Speed , Motor Torque, Armature Voltage and Field Current. - 50 Amp MCCB for control and Isolation of each DC Drive. - Input Ac side Semi-Conductor Fuses and line Reactor for Protection. - Encoder interface card for pulse encoder - Profibus interface card mounted on the drive module for High Speed communication with drive/ PLC. - Block Programming with Readymade function Block to do Co-Ordinated Drive Control. - 16 Character, 2 Line MMI Display – Keypad unit per drive , mounted on the front door of the drive panel for for drive programming and diagnostics. - DOL Starter through MPCB for the Blower fan of Motor. - All the above enclosed in 2 mm CRCA steel Sheet enclosure having IP 31 Protection , with Aluminum Busbars. - Switchgear of Siemens/Schneider/L & T Make	1 Sets
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8. DC Variable speed drive for 55 Kw , 440 VDC, 130 Amp DC motor for Rake

- 1 Set

All the drives and switchgears are suitably rated to take care of Inrush current of 200 % for period of 10 Sec. Field Supply is integrated in the Drive module. Drives are of CLASS IV overload capacity.

A	- Parker DC Variable Speed , 590 series- Digital Microprocessor based Thyristor DC Drive of 270 Amp Continuous Current, suitable for 55 KW, 4QDC Motor for Rake . The Drive operates in closed loop, Speed control mode, to control Speed , Motor Torque, Armature Voltage and Field Current. - 200 Amp MCCB for control and Isolation of each DC Drive. - Input Ac side Semi-Conductor Fuses and line Reactor for Protection. - Encoder interface card for pulse encoder - Profibus interface card mounted on the drive module for High Speed communication with drive/ PLC. - Block Programming with Readymade function Block to do Co-Ordinated Drive Control. - 16 Character, 2 Line MMI Display – Keypad unit per drive , mounted on the front door of the drive panel for for drive programming and diagnostics. - DOL Starter through MPCB for the Blower fan of Motor. - All the above enclosed in 2 mm CRCA steel Sheet enclosure having IP 31 Protection , with Aluminum Busbars. - Switchgear of Siemens/Schneider/L & T Make	1 Sets
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Sl No	Description	Qty
9	Common Incomer	2 Set
a	6300 Amp , EDO type ACB as common incomer	2 Set
b	Energy meter etc on the panel door	2 set
10	Control Desk	1 Set
a	Fabricated Control desk with Push buttons and Meters for the control of Stands, CC shear ,	
b	Siemens Win cc Scada	
c	Siemens 300 Series PLC	
11	Loose Supply Items	1 Set
a	Encoder with Coupling	28 Set
b	Profibus Cable	1 Lot
c	HMD	3 Nos.
d	Proximity Switch	6 No
12	Loop Scanner 2 Nos	1 Set
13	Essential Commissioning Spares	1 Set
a	Control card	1 Set
b	Power card for stack controller	1 No
c	Calibration card for stack controller	1 No
d	Encoder card	1 No
e	Power card for 270 Amp 4 Q Dc drive	1 No
f	Thyristor for 350 Kw stand drive	2 Nos
g	Thyristor for 270 Amp drive	2 Nos
h	Fuse for the stack of stand drive (1000 Amp)	1 No
i	Fuse for the field supply (40 Amp)	2 Nos

